EXPERIMENT 5: SECOND ORDER STEP RESPONSE OF A PARALLEL RLC CIRCUIT

INTRODUCTION

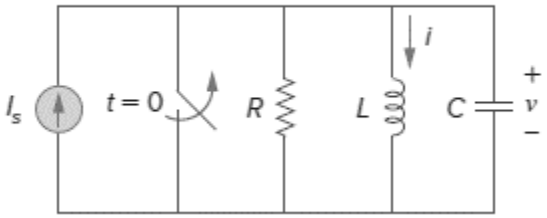


Figure 1: Parallel RLC circuit with an applied DC current

Consider the parallel RLC circuit shown in figure 1, by applying KCL at the top node for t > 0,

$$\frac{v}{R} + i + C\frac{dv}{dt} = I_s \quad \text{.... (1)}$$

But,

$$v = L\frac{di}{dt} \quad \text{....(2)}$$

Therefore the total solution, substitute eqn. (2) into eqn. (1)

$$\frac{d^2i}{dt^2} + \frac{1}{RC}\frac{di}{dt} + \frac{i}{LC} = \frac{I_s}{LC}$$

This solution has two components: the transient response It(t) and the steady-state response Iss(t); that is,

$$i(t) = i_t(t) + i_{ss}(t)$$

The complete response $I_t(t)$ for the overdamped, underdamped, and critically damped cases are:

$$i(t) = I_s + A_1e^{s_1t} + A_2e^{s_2t} \quad \text{(Overdamped)}$$

$$i(t) = I_s + (A_1 + A_2t)e^{-\alpha t} \quad \text{(Critically damped)}$$

$$i(t) = I_s + (A_1\cos\omega_d t + A_2\sin\omega_d t)e^{-\alpha t} \quad \text{(Underdamped)}$$

The constants A_1 and A_2 in each case can be determined from initial conditions for i and di/dt. Keep in mind the transient response equation only applies for finding the inductor current i. but once the inductor current $i_L = i$ is known, we can find $v=L\ di/dt$, which is the same voltage across inductor, capacitor and resistor. Hence, the current through the resistor is $i_R=v/R$, while the capacitor current is $i_c=C\ dv/dt$.

Damping factor, $$\alpha = \frac{1}{2RC}$$

Natural frequency, $$\omega_0 = \frac{1}{\sqrt{LC}}$$

The roots, $$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}, \qquad s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

Damped frequency, $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$

OBJECTIVE

To measure the response of a parallel RLC circuit to an applied DC current and to observe the effect that changes in one parameter, R, have upon the response.

EQUIPMENT AND COMPONENT

1. DC Current source, 24mA
2. Switch
3. Resistor, R (2k Ω, 500Ω, 100Ω)
4. Inductor, L (25mH)
5. Capacitor, C (25nF)
6. Oscilloscope (optional)

PROCEDURE

1. Wire up the circuit as shown in Figure 1. Set the value for Resistor, R = 2k Ω, inductor, L= 25mH, and capacitor, C=25nF. Set the DC Current source to 24mA.
2. Calculate step response parameter and fill in the data table 1.
3. Observe and record the step response curve.
4. Repeat step 1 to 3 by decreasing the value of resistor to 500Ω and 100Ω, and fill in the step response parameter in the data table 2, 3.
5. Plot all these three (3)-response curve for underdamped, overdamped and critically damped on the same graph.

RESULT

Table 1: Underdamped response

Parameter	Calculated value
α	
ω_o	
ω_d	
$S_{1,2}$	
$A_1 \; and \; A_2$	
Eqn. for $i_t(t)$	
$i_t(0)$	
$i_t(\infty)$	
$i_t(0.08ms)$	
$i_t(0.4ms)$	

Table 2: Critically damped response

Parameter	Calculated value
α	
ω_o	
$S_{1,2}$	
$A_1 \; and \; A_2$	
Eqn.for $i_t(t)$	
$i_t(0)$	
$i_t(\infty)$	
$i_t(0.1 \; ms)$	

Table 3: Overdamped response

Parameter	Calculated value
α	
ω_o	
$S_{1,2}$	
$A_1 \; and \; A_2$	
Eqn.for $i_t(t)$	
$i_t(0)$	
$i_t(\infty)$	
$i_t(0.2ms)$	
$i_t(0.3ms)$	

DISCUSSION

1. Elaborate your result and findings in this experiment.
2. Compare the different between these three-step responses.
3. Which response curve are the fastest approaches the step input?
4. Show all your working calculation for Table 1, 2 and Table 3.